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Analysis

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Energy and socioeconomic system transformation through a decade of IPCC-assessed scenarios

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Charting future emissions pathways is a central tenet of IPCC assessment reports (AR), yet it is unclear how underlying drivers (including around policy and technology) have influenced the evolution of emissions pathways. Here we compare scenarios in AR5 and AR6 and find that scenarios without specific climate policies enforced have shifted lower in each scenario generation, owing to falling low-carbon technology costs and reduced expectations for economic growth, reducing fossil-fuel shares in energy and industry. Mitigation pathways consistent with 1.5–2 °C have seen increasing electrification rates and higher shares of variable renewables in electricity in more recent scenario generations, implying reduced reliance on coal, nuclear, bioenergy and carbon capture and storage, reflecting changing costs. Despite the shrinking carbon budget due to insufficient recent climate action, mitigation costs have not increased given more optimistic low-carbon technology cost projections. Moving forward, scenario producers must continually recalibrate to keep abreast of technology, policy and societal developments to remain policy relevant.

The decade in between the Sixth (AR6)[1] and Fifth (AR5)[2] assessment reports of the IPCC has seen dramatic cost reductions and performance improvements in key clean technologies such as solar photovoltaic (PV), battery storage and wind power[3–7]. However, it has also seen relative stagnation in long-considered mitigation 'game-changers'[8], resulting in concerns over the plausibility of the diffusion scales assumed in modelled mitigation pathways—regarding carbon capture and storage (CCS)[9] or carbon dioxide removal (CDR), particularly bioenergy with CCS (BECCS)[10–13]. The same decade saw the introduction of low-carbon policies and targets[14] that have helped shift away from 'baseline' (no-climate-policy) scenarios[15]—alongside increased climate awareness among citizens[16] and disruptive socioeconomic events amidst lingering effects of the Great Recession that include a pandemic, international conflict and large-scale energy crisis.

These developments have had varied implications for climate action, thereby altogether offering useful opportunities to assess the capacity of integrated assessment models (IAMs) to keep up[17] and to understand how the insights emerging from IAM exercises have changed throughout the decade. In this period, the scientific field of IAMs has grown substantially in research output[18] and user interest[19], driven by increasing dedication to the climate emergency[20] and improvements in computational power[21]. This growth has allowed thousands of scenarios projecting mitigation pathways with predefined emissions or climate targets to be published in the literature, revealing insights on how emission reductions might be prioritized across countries and sectors.

To fully reflect on what can be learnt from recent scenario literature and understand how potential developments onwards can

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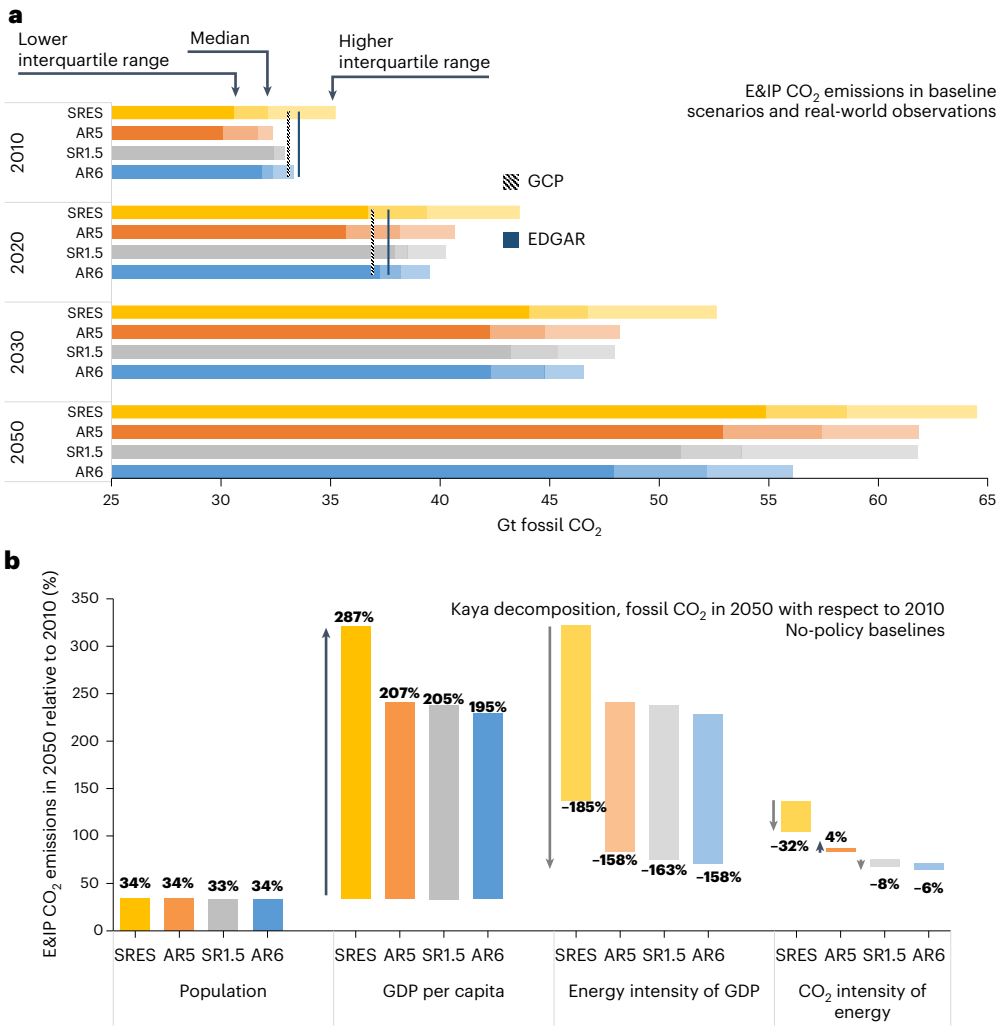

Fig. 1 | Analysis of E&IP CO_2 emissions in baseline scenarios. a, Medians and interquartile ranges of emissions outcomes from no-policy baselines, for IPCC SRES, AR5, SR1.5 and AR6. Values for 2010 and 2020 include real-world observations by Global Carbon Project[53] and EDGAR v.8.0 (ref. 54). For these observations, we took the average of 2009–2011 and of 2019 and 2021 for the value in 2010 and 2020, respectively, to avoid annual variability (including Covid-19 impacts) which most models do not account for. **b**, Kaya identity components for mean emission pathways, separately for SRES, AR5, SR1.5 and AR6. Values for SRES excluded industrial process emissions, and have been multiplied by 1.06 to be comparable with AR5, SR1.5 and AR6 scenarios, which commonly report the sum of E&IP emissions. Statistics are drawn from model means (Methods) and SRES $n = 15$, AR5 $n = 53$, SR1.5 $n = 16$, AR6 $n = 47$. Supplementary Table 1 gives an overview of included scenarios.

be addressed[22], it is instructive to assess what scenario database ensembles say about the evolving role of technology and climate policy[23]. Here, we present such analysis comparing all scenarios assessed within the IPCC reporting process spanning the last decades. The analysis of scenario ensembles spanning four IPCC cycles provides a more dynamic overview of progress, allowing to track the evolution of baseline projections—as a proxy of climate effort effectiveness, improvements in modelling capacity and trends in low-carbon technoeconomic developments—and, in turn, of the solution space provided by IAMs to achieve predefined climate targets. Finally, we aim to understand how mitigation portfolios have evolved in IAMs and the driving factors behind this evolution.

Results

We have gathered, compiled and compared data from IAM scenarios included in four online scenario databases (Extended Data Table 1). These include AR5 published in 2014[24], the Special Report on Global Warming of 1.5 °C (SR1.5) in 2018[25] and AR6 in 2022[26], which all benefit from a harmonized reporting template for all socioeconomic, energy system, emissions and climate outcomes, thereby allowing for systematic comparison of several dimensions of mitigation. We additionally include inputs from the Special Report on Emissions Scenarios (SRES), which was used to inform the AR3 and AR4 (ref. 27). As baseline-only scenarios, SRES are compared exclusively to baseline scenarios across ensembles. All observations throughout this section are 'model means'—that is, the mean indicator value from all scenarios of similar climate ambition from the same model (Methods), guaranteeing results are not biased towards specific models with a large number of scenarios in IPCC assessments.

Evolution of baseline scenarios

Baseline scenarios are an important reference point. First, they are unconstrained by carbon budgets and mostly driven by assumed continuation of current technology, economic and societal trends. Analysing baseline scenarios, therefore, is important to understand the significance of the observed differences in more ambitious scenarios. Second, the transformative potential of a project or political instrument has been argued to correspond to its potential to shift from an assumed business-as-usual (BAU)[28]. BAU scenarios should reflect the expected change under socioeconomic assumptions,

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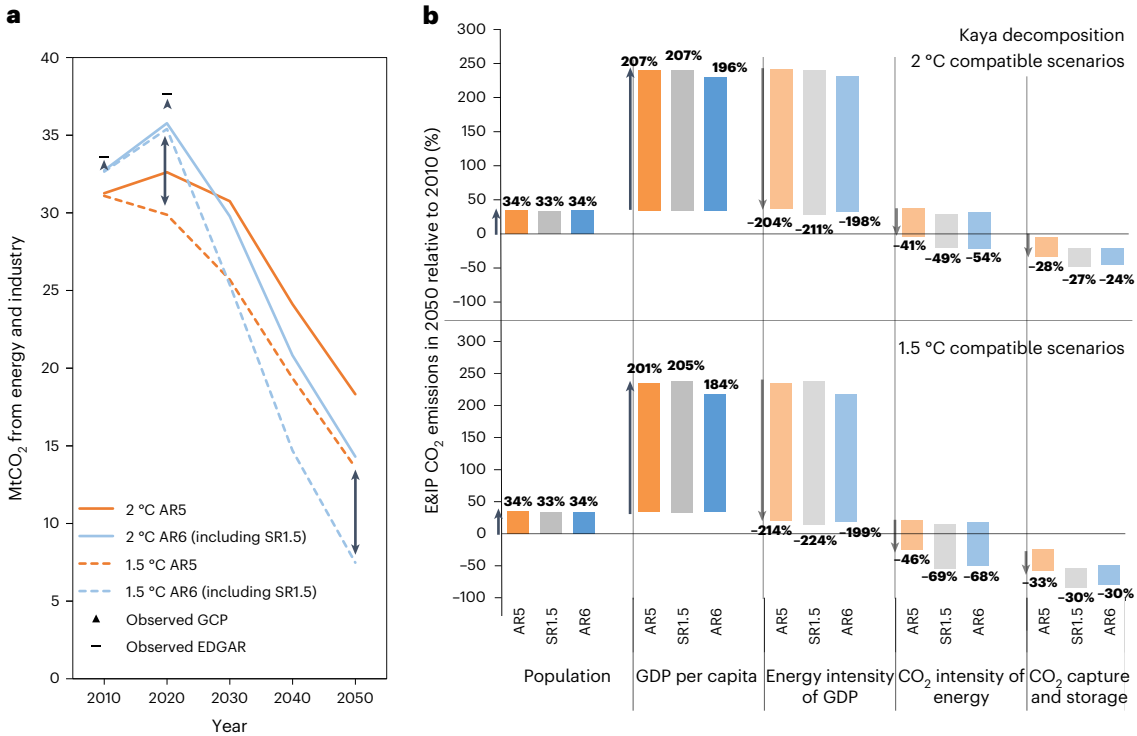

Fig. 2 | Analysis of E&IP CO_2 emissions in mitigation pathways. a, Mean pathways in AR5 and AR6 from 2010 to 2050 and observed emissions in 2010 and 2020 (Fig. 1). Arrows in 2020 and 2050 reflect how the delayed mitigation in AR6 scenarios relative to AR5 scenarios leads to deeper required reductions in 2050 in AR6 scenarios to be compatible with the same climate target. Values for 2010 and 2020 include real-world observations by Global Carbon Project[53] and EDGAR v.8.0 (ref. 54). **b**, Kaya identity components for 2 and 1.5 °C compatible scenarios, separately for AR5, SR1.5 and AR6. Statistics are drawn from model means (Methods) and Supplementary Table 1 gives an overview of included scenarios for each indicator.

whereas mitigation scenarios outline more ambitious transformations of sociotechnical and economic systems towards meeting ambitious climate goals.

Baseline scenario projections for CO_2 emissions from energy and industrial processes (E&IP) in 2030 and 2050 have gradually decreased from SRES towards AR6 (Fig. 1a), reflecting transformation that has gradually happened (for example, due to policy efforts and low-carbon technology deployment in the historical years of models), evolution of modelling assumptions and/or capabilities (for example, relative costs and enhanced model representation of low-carbon technologies) or changes in assumed socioeconomic trajectories. Comparing scenario emissions in 2010 and 2020 with historical values shows that 2010 emissions were generally underestimated (below median) and 2020 emissions (2019–2021 average to largely avoid Covid-19 effects) overestimated in all datasets. Logically, older datasets show higher variability, as estimations are further away from model base years, allowing initial differences to increase over time. The improved baseline starting point, even in the absence of strong climate policies, and the clear overestimation of 2020 emissions, is compatible with recent literature pointing to futures of more than 4–5 °C end-of-century warming being increasingly unlikely[15,29].

Disentangling the 2010–2050 evolution of mean emission pathways of no-climate-policy baselines (Fig. 1b) in Kaya identity components (Methods) shows how differences in mean E&IP CO_2 emissions (Fig. 1) are driven by a combination of causes. First, relative to AR5 and AR6, SRES projected significantly higher gross domestic product (GDP) growth, but also stronger decline in energy intensity of GDP and CO_2 intensity of energy, even in the absence of policy. The baseline emission reduction from AR5 to SR1.5 and AR6 is driven by a combination of all Kaya components, with again lower projected GDP growth playing a significant role, being the main driver behind AR5 scenarios overestimating emissions in 2020[30]. Overall, however, this overestimation of future GDP growth throughout the modelling ensembles may be driven by long-term uncertainty or structural assumptions in projections[31] (Extended Data Fig. 1).

Evolution of mitigation pathways

When giving IAMs the objective to limit global temperature increase, radiative forcing or cumulative emissions to certain levels, they typically estimate the cost- or welfare-optimal path over time and space towards that level[32]. Nevertheless, model users determine from which year onwards IAMs must look for this path. AR5 scenarios assumed a level of mitigation activity by 2020 that did not eventuate, with the SR1.5–AR6 ensembles therefore deferring the onset of substantive mitigation activities until post-2020. The relatively delayed and higher peak in emissions resulted in deeper required emission cuts in 2050 in SR1.5–AR6 to be compatible with <1.5 °C warming (Fig. 2a). A similar but more moderate effect can be observed for 2 °C pathways, a pattern repeated for every region (Extended Data Fig. 2), confirming that it is a result of models attempting to find the optimal pathway to specified targets, rather than underlying region-specific developments.

In terms of Kaya components (Methods), SR1.5 and AR6 include a much more substantial reduction in CO_2 intensity of energy supply, than does AR5 (Fig. 2b), suggesting that these two ensembles rely heavily on supply-side technology innovation to compensate for the lagging mitigation response of the world. This observation may confirm criticisms of IPCC scenarios often relying on largely unprecedented technological change rather than slower economic growth or demand reduction for reaching climate objectives[33], although there is also an incremental role for slower economic growth projections (AR6) or faster declining energy intensity of GDP (SR1.5) to make up for this delay relative to AR5. Interestingly, the role of technological carbon

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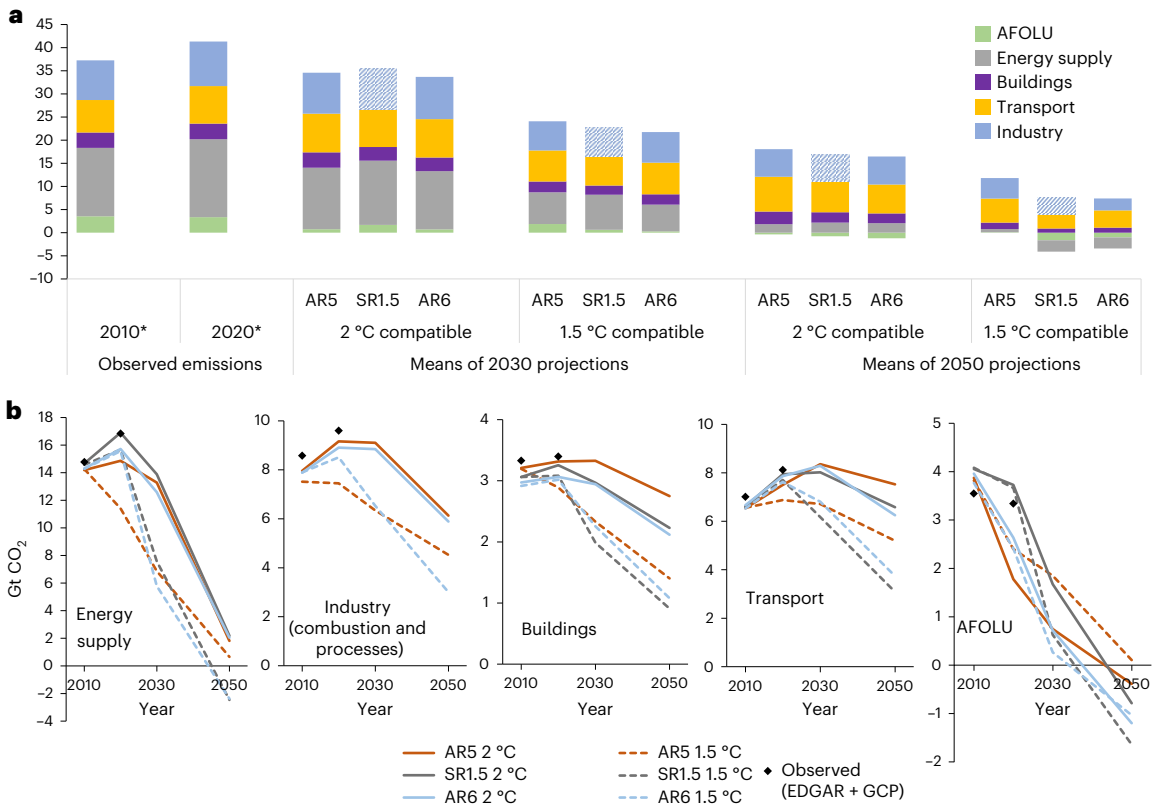

Fig. 3 | Global CO_2 emissions by sector. Mean values for sectoral contribution of CO_2 emissions in 2010 to 2050, separately for scenarios that are 2 and 1.5 °C compatible and for scenarios in IPCC AR5, SR1.5 and AR6. **a**,**b**, Absolute distribution per scenario (**a**) and evolution over time by sector (**b**). The asterisks indicate averages of observed values from EDGAR v.8.0 (ref. 54) (energy consumption sectors) and GCB[53] (AFOLU sector) for 2009–2011 (as estimate for 2010) and of 2019 and 2021 (as estimate for 2020) to avoid annual variability (including Covid-19 impacts) which most models do not account for. Industry estimates from SR1.5 omitted because of inconsistent reporting of emissions from industrial processes. Statistics are drawn from model means (Methods) and Supplementary Table 1 gives an overview of included scenarios for each indicator.

removals (such as CCS) slightly decreased from AR5 to AR6, despite higher mitigation pressure.

The sectoral stocktake in mitigation efforts in 2010–2050 shows that the largest absolute contributions come from energy supply (covering all transformation stages—for example, mining, refining and electricity), but additional effort in moving from 2 to 1.5 °C pathways comes mostly from industry, buildings and transport (Fig. 3). Across scenario ensembles, there is a shift towards deeper mitigation in end-use sectors towards 2050, post-AR5, both for more modest 2 °C scenarios and more clearly for 1.5 °C scenarios, reflecting improved model incorporation of end-use technologies, such as electrification and hydrogen in industrial processes, battery-electric vehicles in transportation and electric heat pumps in buildings and/or the maturity of these technologies (for example, lower costs or growth constraints) post-2015[34,35].

The evolution of CO_2 emissions from Energy Supply and AFOLU varies less in absolute quantity between scenario databases, but notably these sectors go towards net-negative emissions in SR1.5–AR6, instead of carbon neutrality (AR5). Energy-sector CO_2 emissions become negative primarily as a result of electricity generation and fuel supply by BECCS, which first must offset residual sectoral emissions before becoming net-negative. Lower residual emissions in electricity generation (for example, more renewables) can facilitate lower emissions in the energy sector even if BECCS use is moderated. AFOLU CO_2 emissions represent a balance between reforestation, afforestation, deforestation and—depending on model—forestry. There is a trade-off between BECCS and AFOLU because of limited land availability and saturation of AFOLU over time[36].

Energy technology mix and macroeconomic costs

Focusing on mitigation scenarios compatible with 2 or 1.5 °C, results indicate gradual evolution of mitigation portfolios. In terms of primary energy mix (Fig. 4a), coal shares have dropped in more recent ensembles: while coal still played a non-negligible role in deep mitigation scenarios in AR5 (~10% of primary energy by 2050), its share is <10% in all 2 °C/1.5 °C scenarios in AR6. Reliance on natural gas was relatively stable among scenarios in AR5 (~20–25% of primary energy) but decreased in 1.5 °C scenarios in SR1.5–AR6 (towards ~15%). This drop is probably related to the need for deeper mitigation relative to AR5 (Fig. 2). Reliance on bioenergy for mitigation is also slightly lower in more recent scenarios, whereas reliance on solar and wind in the most ambitious scenarios increased from 10–15% (AR5) to 20–45% of total primary energy demand by 2050 (AR6), in response to rapidly declining technology costs and enhanced model representation of variable renewables (VRE) integration in the electricity system (for example, coupled with batteries). This last effect is clearly related to the electricity mix and significance in final energy demand, due to enhanced electrification (Fig. 4b). For example, the share of renewables in electricity increases from 40–50% (AR5) to 60–80% (SR1.5–AR6) by 2050 throughout scenarios, while nuclear drops from 15–20% to 10–15%, probably related with cost escalation and limited political support after the 2012 Fukushima event[37]. The share of electricity in final energy demand increases with deeper mitigation, but the overall level is ~10% higher in AR6 over AR5, leading to electrification rates of ~50% rather than ~40% for the most ambitious scenarios. Total primary and final energy use, both decreasing with increasing mitigation effort, have not significantly evolved since AR5 (Extended Data Fig. 3).

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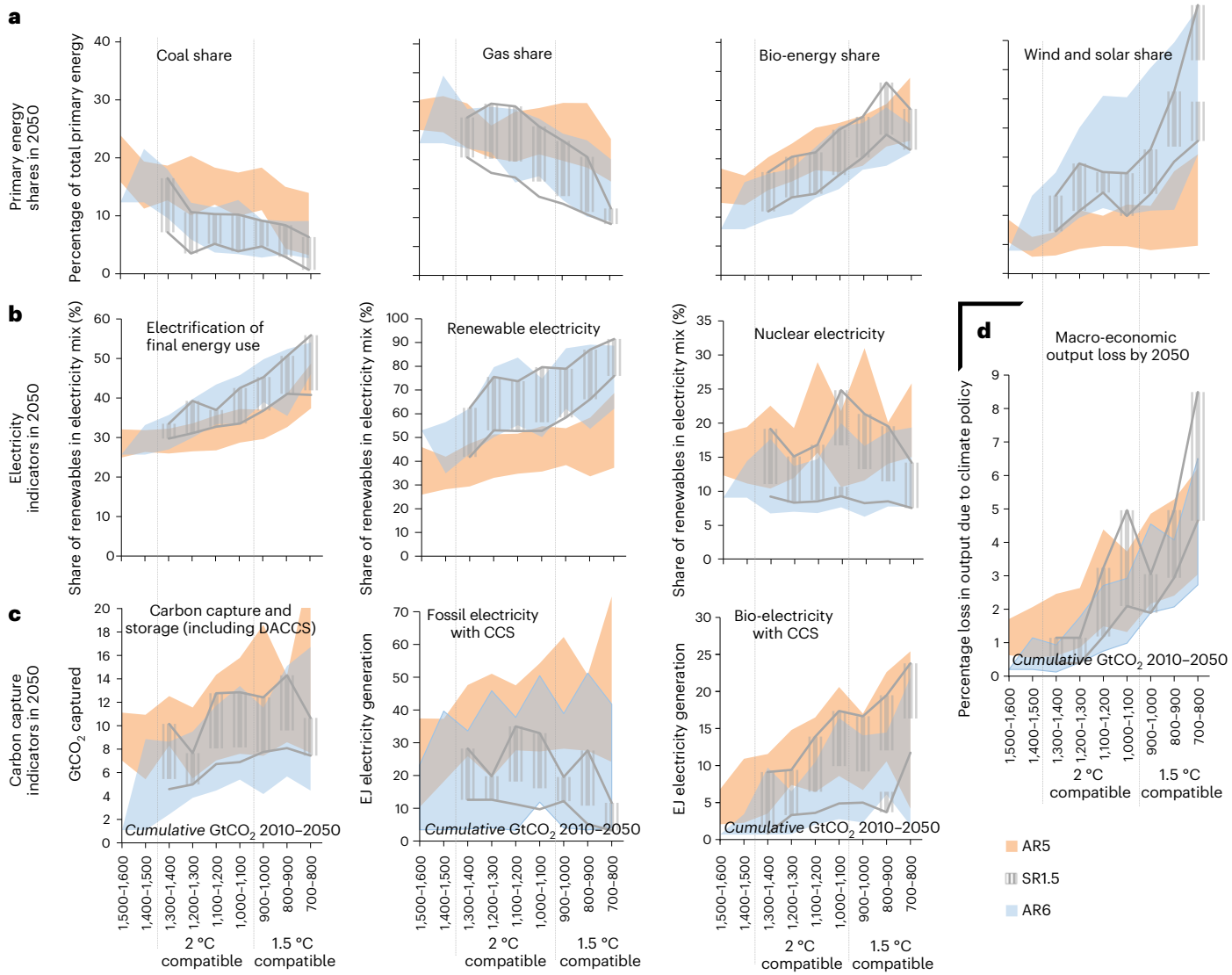

Fig. 4 | Energy system and mitigation indicators by 2050. a–d, Interquartile ranges of structural indicators (primary energy shares (**a**), electricity indicators (**b**), carbon capture indicators (**c**) and macroeconomic output loss in (**d**)) of mitigation pathways by 2050, plotted separately for IPCC AR5, SR1.5 and AR6 as a function of mitigation effort in terms of cumulative fossil CO_2 emissions from 2010 to 2050, marked by 2 °C and 1.5 °C compatibility. The metric for macroeconomic output loss (**d**) differs between reported scenarios, and includes GDP losses, consumption losses or the area under the MAC curve. Statistics are drawn from model means (Methods) and Supplementary Table 1 gives an overview of included scenarios. EJ, exajoule.

Generally, mitigation scenarios have evolved to depend less on CCS (Fig. 4c). The comparison of different CCS elements between scenario ensembles is not straightforward because of changes in the reporting structure of carbon sequestration, but the figure shows that reliance on CCS-powered fossil electricity significantly decreased from AR5 to SR1.5, staying relatively low but with high model variance in AR6, which is compatible with the abovementioned reduced reliance on fossil fuels and non-renewable electricity. Nevertheless, the overall role of CDR has not markedly reduced since AR5. While the role of BECCS marginally decreased, there has been a rise in new CDR technologies such as DACCS[38] (Extended Data Fig. 4). This distinction between CDR technologies is important, as it shows that, while mitigation has shifted towards renewables, the need for carbon removals has not declined with the need for deeper mitigation in SR1.5–AR6.

Interestingly, SR1.5 scenarios appear to coincide more with AR6 trajectories for nearly all variables. The fact that 1.5 °C scenarios in SR1.5, as in AR6, required deeper mitigation compared to AR5 (Fig. 2), may play a role. However, for many variables (coal and nuclear shares, renewable electricity and electrification), the resemblance clearly starts with 2 °C scenarios already, and the similar tendency of 2 °C scenarios throughout the IPCC scenario ensembles (Fig. 2) suggests that there are more structural causalities too; for example, updated model structures and/or assumptions.

For decades now, the IPCC has been tasked with assessing the literature on macroeconomic implications of reducing emissions and has responded by publishing both estimates of, and limitations inherent in, long-term macroeconomic projections[39]. Acknowledging the numerous methodological issues in these estimates, as well as inherent differences in how models address the economics of mitigation, we track how assessments of macroeconomic consequences have evolved from AR5 to AR6 (Fig. 4d). We observe that, for 2 °C scenarios, estimates have shifted from reductions of 2–3% of economic activity (GDP) in AR5 to 1–2% in AR6. For 1.5 °C scenarios, these differences are smaller, dropping from 3–5% to 2–4% of total GDP. Nevertheless, considering the deeper mitigation requirements in 2050 in AR6 relative to AR5 (Fig. 2), the observation that macroeconomic costs are lower in AR6 signals that a broader suite of mitigation options was

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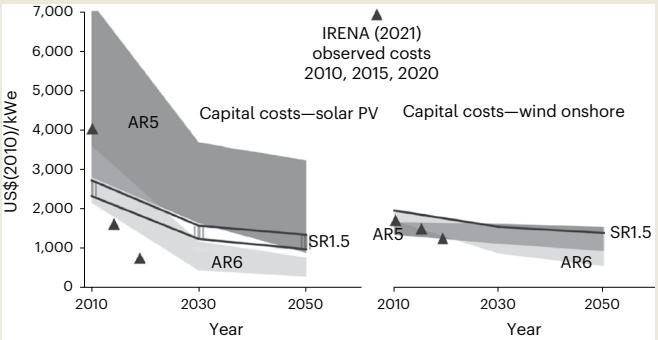
BOX 1

The role of technology in evolving mitigation scenarios

Several clean energy technologies have been scaled up significantly over the past decade, with cost reduction and performance improvements unforeseen at this scale in modelled scenarios, which have been gradually implemented in more recent scenarios. The clearest example is solar PV, for which the cost assumption for 2020 was three to six times lower in AR6 than in AR5 (interquartile ranges), and observed costs around 2020 according to IRENA (2021)[55] were lower than assumed costs for 2050 in AR5 and SR1.5 scenarios. Similarly, for technologies such as onshore and offshore wind, battery storage, electric vehicles, hydrogen electrolysers and electric heat pumps, recent cost reductions and/or performance improvements comfortably exceeded modelled assumptions in earlier model runs and IPCC reports. The updated representation of cheaper-than-expected low-carbon technologies in models causes projected baseline emissions to be lower, as clean technologies compete against dirty alternatives even in the absence of climate policy (Fig. 1), and mitigation technologies to shift radically with the same mitigation target (Fig. 4).



Moreover, the representation of end-use sector technologies has been substantially improved from AR5 towards AR6. This continuous evolution of IAMs also opens new opportunities to mitigation[43,47], and can be expected to continue into future scenario exercises. Recent industry-specific modelling exercises with higher sectoral detail achieve larger emission reductions than IAM scenarios considered in the IPCC cycles assessed in this article[56,57], to the point that some sectors previously considered as hard to abate may now have the option of a much steeper mitigation trajectory, such as the iron and steel sector with new steel-related investment in the European Union (EU) already dominated by low-carbon options (for example, electric arc furnace and hydrogen-based direct reduction of iron). We expect that these recent developments will also be reflected in the new generation of IAM scenarios.

BOX 2

Scenarios from European modelling teams dominate IPCC assessed scenario database, shaping energy transition patterns

The changing patterns we found in the indicators examined in this study (Fig. 4) were especially accentuated in scenarios developed by organizations from Europe: AR6 scenarios from European organizations had a higher share of solar PV and wind power in electricity generation and primary energy as well as higher electrification than the rest—and less reliance on nuclear and CCS (Supplementary Fig. 1). These differences were less pronounced in AR5 and SR1.5 and were even reversed for bioenergy. Similar patterns were found by comparing scenarios of different warming levels between European and other organizations (Supplementary Figs. 2–4), with 2 °C scenarios showing the largest differences. While scenario developers from Europe were also using lower cost assumptions in AR6 for solar and wind than the rest (Supplementary Fig. 5), technological costs did not seem to be the only reason for these differences. For instance, even when controlling for the cost assumptions of solar PV and wind power, the share of renewables in primary energy (Supplementary Fig. 1) was still statistically significantly higher in scenarios coming from European organizations than the rest (median of 25% versus 10%, one-way analysis of covariance, $F = 56.49$, $P < 0.001$). Similarly, ref. 3 showed that projected solar PV growth was higher in AR5 and SR1.5 scenarios from European organizations compared to scenarios from Asian or North American organizations.

European scenario developers have contributed most scenarios to the examined reports (71% in AR6, 75% in SR1.5 and 66% in AR5; Supplementary Table 3), so they expectedly have a large impact on the overall patterns in the IPCC scenario databases. Since their share in the databases has not changed markedly from AR5, the significant evolution observed here may be due to an increased use of scenario designs that explore higher shares of renewables and/or feasibility concerns[58].

available in AR6, assumptions over future mitigation technology costs have become more optimistic and/or structural changes in IAMs have lowered the resistance to major shifts in energy systems. Interestingly, the macroeconomic implications in SR1.5 largely resembled those of AR6 for 2 °C scenarios, but those of AR5 for 1.5 °C scenarios. This may signal that assumptions about major energy transformations may have been more optimistic in SR1.5 relative to AR5, but model capabilities for reaching deep mitigation were not yet as well-developed as in AR6. Note that these costs do not include climate impacts, and the economic impacts of climate inaction as outlined in IPCC WGII may well surpass the IAM-reported costs of mitigation[39].

The figures in this study show a gradually evolving emission/temperature future (in no-climate-policy baselines) and transformation path for reaching 2 °C and 1.5 °C futures, as well as increasing mitigation stringency required for nearly all sectors and regions to achieve 1.5 °C. We discuss likely technological drivers (Box 1) and political funding-related drivers (Box 2) for these changes.

Discussion

IAM research has seen an impressive uplift during the past two decades. Combined with consistent scenario reporting[40], this allows for a multidimensional tracking of how transformation pathways have evolved. Our analysis takes aim at this evolution over the last decade through a detailed analysis of thousands of IAM scenarios in IPCC scenario databases.

There is continuous evolution of most analysed indicators over scenario ensembles with a clear direction towards increasing

electrification rates and higher VRE shares in electricity generation. Simultaneously, the use of nuclear, bioenergy and CCS with fossil energy has declined in more recent scenarios, despite more stringent 2050 targets. Since 2010, remaining carbon budgets have shrunk by ~400 $GtCO_2$ (~0.5 °C), yet the achievability and macroeconomic implications of delivering the same climate target have not markedly changed. An important interpretation of this is that AR5 portrayed more ambitious climate action as harder and costlier than nowadays calculated. The SR1.5–AR6 scenario ensembles feature more ambitious emissions cuts in energy end-use sectors (industry, buildings and transport), while allowing energy supply and AFOLU to reach net-negative CO_2 emissions.

The main causes we found behind the explored differences between scenario ensembles also imply lessons for future scenario development. A major driver lies in faster-than-expected reduction in clean technology costs and improved representation of critical end-use technologies (Box 1). The deviation between projected and observed trends in many key technologies[41,42] points to the need for sustained effort to further update models. Any failure by IPCC scenarios to adequately reflect the practicability/plausibility of certain changes becomes more significant with declining carbon budgets. It is, therefore, important that modelling communities work closely with stakeholders to keep their critical technology assumptions up-to-date amidst rapidly evolving market dynamics. Since CDR levels in mitigation scenarios are dependent on the emission reduction levers in hard-to-abate sectors included in models[43], such developments may reduce reliance on CDR.

A second cause we identified was political (Box 2): narratives stemming from a European context may be more pro-renewables given the rapid roll-out of solar and wind envisioned in the European Green Deal[44] and, perhaps, the EU being a major funder of such studies (for example, via H2020 and Horizon Europe; Supplementary Table 3)[18]. The dominance of scenarios from European institutes in scenario ensembles may explain the increased focus on VRE in more recent scenario ensembles. European narratives may, however, contrast with narratives from developing economies, such as India, where mitigation strategies are interwoven with socioeconomic development[45].

High economic growth projections have historically been used as an argument to delay mitigation, as future generations would be more capable of paying for adaptation and/or CDR technologies to regulate temperatures[46]. Although most scenarios in the explored ensembles are not based on intergenerational cost–benefit analysis, the observation that projections for future economic growth have been continuously corrected downwards (Fig. 1 and Extended Data Fig. 1) may imply reduced innovation potential for breakthroughs and value added of CDR technologies[31]. Such technologies, therefore, increase future costs of both mitigation (through later action) and adaptation (through temperature overshoot)[47], while also involving other risks to future societies[13]. The track record of real-world technology costs outperforming forecasts (Box 1) illustrates that lacking forecasts of inputs to mitigation scenarios have falsely favoured delayed action throughout the past decade.

However tempting to conclude that renewables in future low-carbon electricity projections will continue to increase, owing to spectacular cost reductions and deployment relative to CCS and nuclear, this is not certain. Grid balancing, social acceptance, mineral supply chains and so on must be considered as potential offsets to continued capital-cost declines. The future direction of other observed trends is also unclear; notably, although projected mitigation costs have fallen (or, for 1.5 °C pathways, largely remained unchanged) with decreasing technology costs more-than-offsetting shrinking carbon budgets, this balance can only last so long. Inevitably, especially for 1.5 °C pathways, mitigation costs will rise, as the remaining carbon budget eventually diminishes and emissions removals become central to keeping below this temperature threshold. A clearer trend, from AR5 to AR6, is the reduction in coal and gas shares in future primary energy demand in the lowest carbon budget scenarios. This could well continue into AR7 but, regardless, near-term geopolitical developments must be factored in—for example, potential lock—into LNG infrastructure in light of a wave of new global investment[48].

It is likely that the 1.5 °C temperature stabilization goal will be exceeded within the next 5–10 years[49], dictating a reframing of scenarios and IAMs. Instead of more elaborate ways to keep below 1.5 °C, focus may shift towards feasibility[50], requiring more critical thinking on how different technologies may be deployed under different policy and regional contexts. This emphasizes the importance of using multiple streams of evidence when using/interpreting scenarios, such as sectoral and national expertise. Models and scenarios are enticing because of their ability to integrate hundreds or thousands of assumptions, but this also makes them susceptible to providing a false sense of confidence in results. The surge of the 'net-zero' concept, stemming from physical climate science[51], and increasingly applied in political pledges and scenario design[50], had no role in AR5, a relatively small role in AR6, but will probably be instrumental towards AR7, by itself conditioning several indicators explored in this study.

The world now has a multitude of mitigation policies in place and, despite lack of a strong tradition of such analysis, their impact will complicate the development of realistic 'reference scenarios' in future exercises. First, without real-world comparison, evaluating no-climate-policy baselines is not as meaningful at the global level[15]. Second, despite a growing number of relevant studies[14,50,52], pathways reflecting current policies and/or Nationally Determined Contributions are also hard to evaluate, since policy constantly changes. Third, backcasting scenarios typically performed to determine how the world could transform and achieve ambitious climate targets mostly apply idealized assumptions, such as globally harmonized carbon prices across sectors/regions, a situation increasingly unlikely to be realized. Evaluating scenarios will, thus, require an increasing degree of expert judgement to elicit policy relevant recommendations.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41558-024-02198-6.

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Methods

For the analysis of climate change mitigation scenarios, we analysed all scenarios with cumulative CO_2 emissions <1,600 $GtCO_2$ for the 2010–2050 period, which includes all scenarios staying below 2 °C temperature increase with at least 50% probability (IPCC, 2021, Table SPM.2)[59]. We categorized them by publication year: roughly, AR5 covers scenarios published in the 2010–2014 period, SR1.5 in 2015–2018 and AR6 in 2019–2021. All scenarios included in the SR1.5 database are excluded from the AR6 to avoid duplication. We compare means, medians and interquartile ranges, specific for the defined time points (for example, 2010, 2030 and 2050, abstracted from the values taken from the databases at for those years), of various key indicators for climate mitigation. In the sections analysing the sectoral stocktake in mitigation and the evolution of Kaya identity components, we focus at mean values from all scenarios, separately for scenarios compatible with 2 and 1.5 °C. In this analysis, we look at cumulative emissions from 2010 to 2050 to order scenarios in the 2 or 1.5 °C category, where applicable. The margins assumed for 2 and 1.5 °C compatibility are, respectively, 1,000–1,400 and <1,000 $GtCO_2$. In section 'Energy technology mix and macroeconomic costs', we plot interquartile ranges as a function of cumulative 2010–2050 global E&IP CO_2 emissions.

The contribution of models to scenario numbers in the databases is skewed, with nearly 40% of scenarios coming from 10% of the most used model versions in both AR5 and AR6 (Supplementary Fig. 6). This relative domination of the solution space by some specific models affects the means and interquartile ranges that are abstracted in this analysis[60]. While in some cases, model dominance correlates with relevance, since modelling teams collaborating in multiple modelling studies will probably generate more impact than those that collaborate in less, this is certainly not always the case. For example, some modelling exercises consist of producing multiple scenarios with marginal differences between them (for example, in ENGAGE[58]), while other modelling exercises consist of producing fewer scenarios with stronger differences between them to illustrate some specified narratives or pathways (for example, the McKinsey scenarios[61]). These latter modelling exercises, however, are not necessarily less influential or produced with less effort. Therefore, and since the purpose of this analysis is to track how the overall mitigation narrative has evolved between different scenario ensembles, the figures in this analysis are based on means and interquartile ranges of unique model versions, that is by taking means of all scenarios from the same model version and same E&IP CO_2 emission level, where this applies (for example, by 2 °C and 1.5 °C compatibility in Figs. 2 and 3 or by cumulative emission increment in Fig. 4), before calculating the figure outputs. For example, if we observe ten scenarios from model A and two scenarios from model B to be in a specific emission range (for example, 2 °C compatibility), we first take the mean of those ten and two scenarios from, respectively, model A and B, and then use the mean, median and/or interquartile range based on those precalculated model means. This way we intend to provide a picture of how scenarios have evolved over time that is not biased by scenario quantity from each model, but rather informed by model heterogeneity. In case of version updates for the same model (for example, model A v.1.1 versus model A v.1.2), they are treated as different models, so as to reflect the potential impacts of model updates on the outcomes. In Supplementary Table 1, we provide numbers for the observations for each analysed indicator. Here, a 1 implies a scenario reported this indicator and 0 implies that it did not.

Apart from a model bias, statistical operations from large scenario ensembles may also suffer from a scenario bias, for example, through an uneven use of potential societal and climate futures[62]. In the case of the shared socioeconomic pathways (SSPs) for example, the majority of scenarios applying the SSP framework in the SR1.5 and AR6 ensembles apply the SSP2 'middle-of-the-road' narrative (Supplementary Fig. 10). A changing balance in narrative use between scenario ensembles may create a bias when analysing and comparing such ensembles[63]. However, the continuous change of 'scenario families' over the scenario ensembles analysed in this study (for example, A1/A2/B1/B2 in SRES, SSPs in SR1.5 and AR6) makes it impossible to account for such a scenario bias in the statistical analysis of different ensembles.

The Kaya identity[64] describes the factors of environmental impact. In this analysis, the Kaya identity is expressed in the form:

$$\text{Baseline scenarios}: F = P \times \frac{G}{P} \times \frac{E}{G} \times \frac{F}{E}$$

$$\text{Mitigation scenarios}: F = P \times \frac{G}{P} \times \frac{E}{G} \times \frac{F_{NA}}{E} \times \frac{F}{F_{NA}}$$

where:

F is global CO_2 emissions from human sources, after CCS abatement;
P is global population;
G is world GDP;
E is global energy consumption;
F_{NA} is global CO_2 emissions from human sources, before CCS abatement (NA, non-abated);
G/P is the GDP per capita;
E/G is the energy intensity of the GDP;
F/E is the emission intensity of energy;
F_{NA}/E is the emission intensity of energy, in the absence of CCS abatement; and
F/F_{NA} is the relative contribution of CCS abatement.

Each of the factors described is estimated separately on the basis of the data availability; for example, depending on the specific scenario reporting. As with other indicators, these are calculated in terms of model means, and are treated as five separate indicators in Supplementary Information Section 2.

Finally, we examined various factors behind the evolution of the scenarios such as evolving technological costs, policy changes and the characteristics of scenario developing organizations. On the last, we mapped each model to the organization that was using it to examine the impacts of characteristics such as the region of origin and the type of organization (figures in Supplementary Information Figs. 1–5, and interpretation in the main text). To examine the role of funders in the priorities of scenario developers, we also extracted funding details for each global scenario study based on the study details and DOI provided in the meta data of each scenario ensemble.

Data availability

The different IPCC datasets used in this study are publicly available from https://www.cger.nies.go.jp/db/scenario/index.html (SRES), https://secure.iiasa.ac.at/web-apps/ene/AR5DB/ (AR5), https://data.ene.iiasa.ac.at/iamc-1.5c-explorer/ (SR1.5) and https://data.ece.iiasa.ac.at/ar6/ (AR6).

Code availability

The full R and Jupyter Notebook codes used to analyse the data in this Analysis is publicly available via Zenodo at https://doi.org/10.5281/zenodo.13998470 (ref. 65).

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Author contributions

D.J.V., S.M., A.G., L.H., P.F., W.O. and G.P.P. coordinated the study design. D.J.V., S.M. and G.X. were responsible for the compilation of the analysis and figures. D.J.V. coordinated the conception and writing of the paper with notable contributions from A.N., G.X., A.G. and G.P.P. and feedback and contributions from all other authors (S.M., L.H., P.F., W.O., M.G.-E., F.F., I.S. and L.C.).

Competing interests

The authors declare no competing interests.

Additional information

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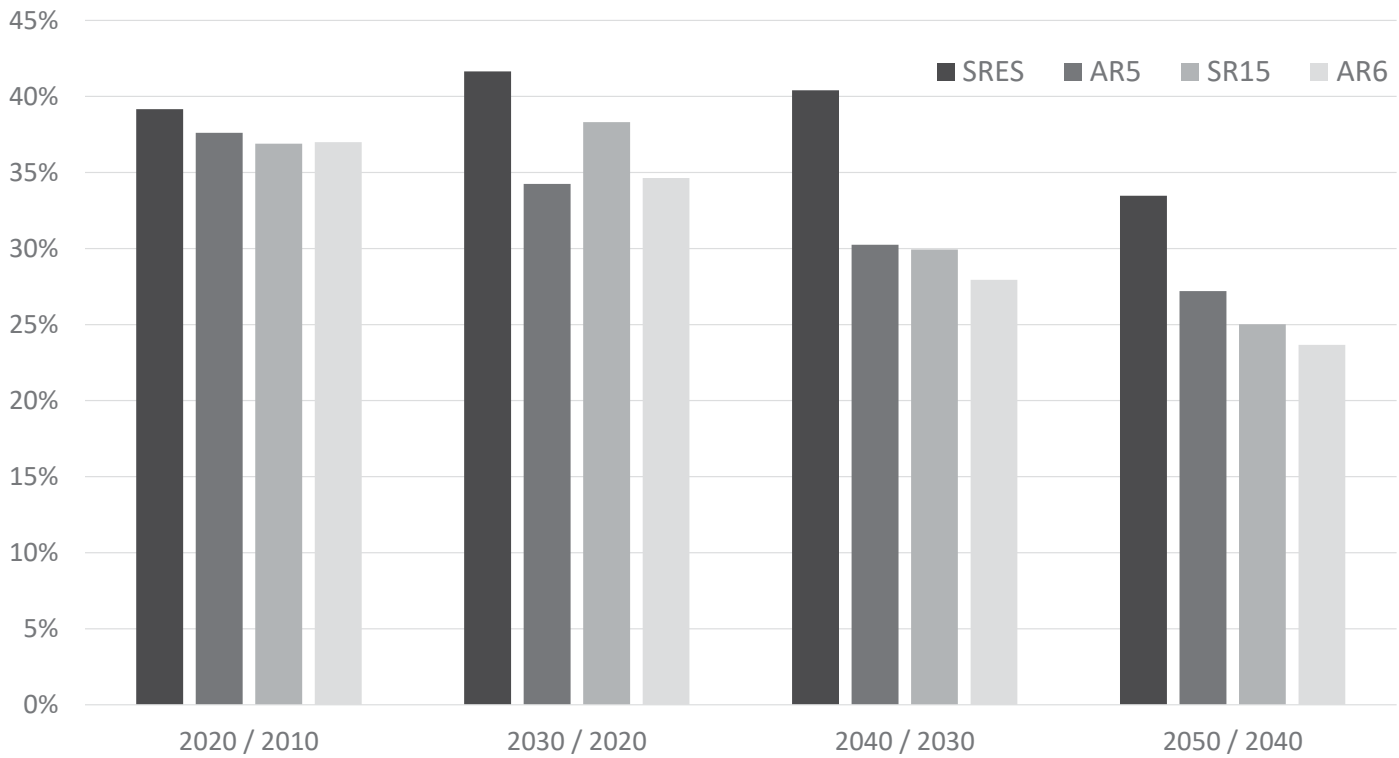

Extended Data Fig. 1 | Assumed GDP growth. GDP growth by decade in identified baseline scenarios in the different scenario ensembles.

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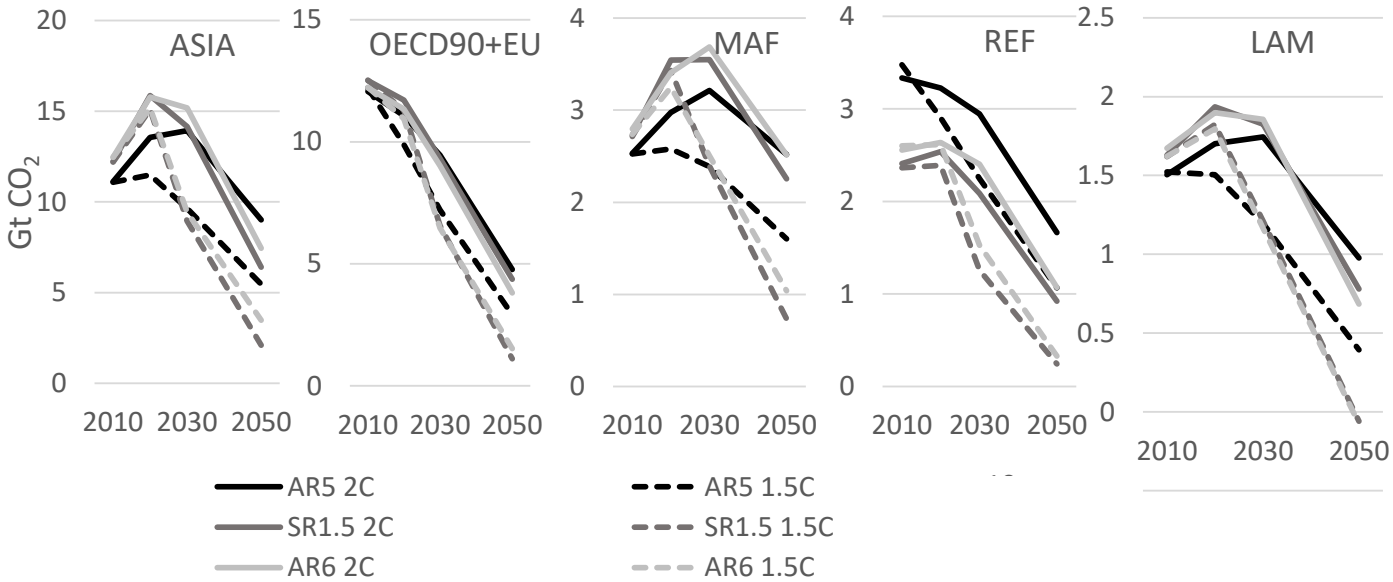

Extended Data Fig. 2 | Regional mitigation scenarios. Mean values for regional contribution of CO2 emissions in 2010 to 2050, separately for 2 °C and 1.5 °C compatible scenarios, and for scenarios in IPCC AR5, SR1.5 and AR6. Statistics are drawn from model-means (see Methods) and see SI-2 for overview of included scenarios for each indicator.

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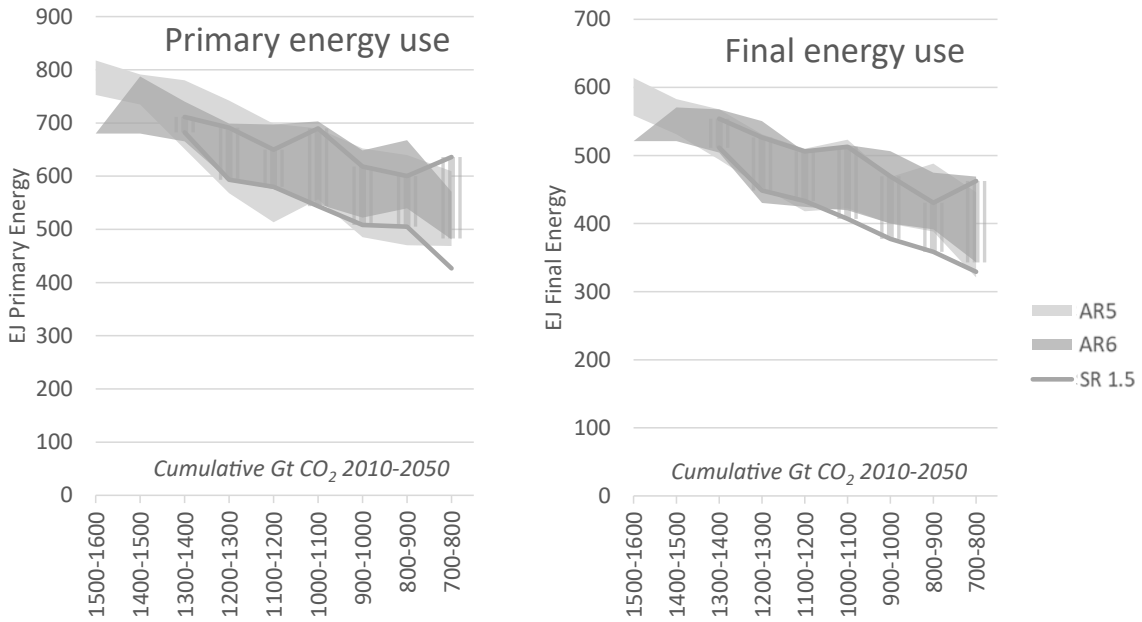

Extended Data Fig. 3 | Primary and Final energy use. Interquartile ranges of structural indicators of mitigation pathways by 2050, plotted separately for IPCC AR5, SR1.5 and AR6 as a function of mitigation effort in terms of cumulative fossil CO2 emissions from 2010 to 2050, marked by 2 °C and 1.5 °C compatibility. Statistics are drawn from model-means (see Methods) and see SI-2 for overview of included scenarios for each indicator.

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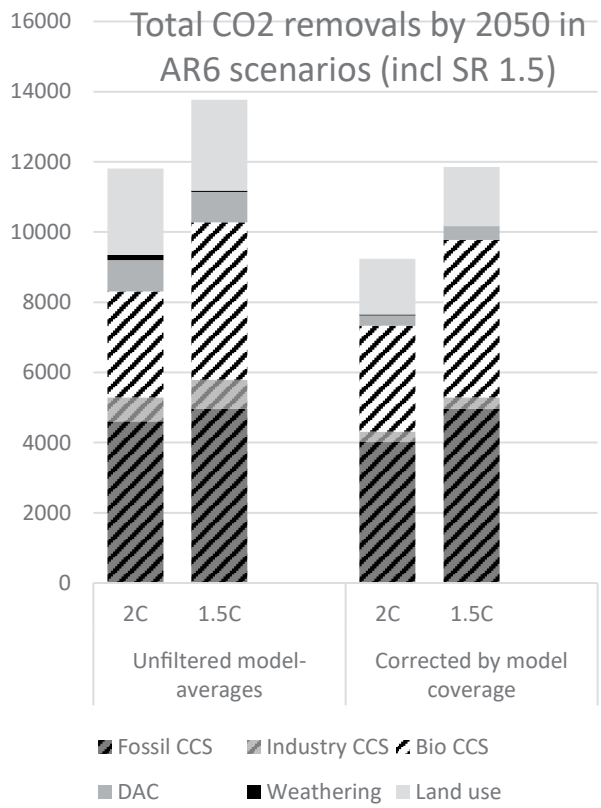

Extended Data Fig. 4 | Total CO2 removals by 2050 in AR6 scenarios (incl SR 1.5). Carbon sequestration in 2050 by method in AR6 scenarios (including SR 1.5) for 2 °C and 1.5 °C compatible scenarios. Since model coverage of more novel sequestration methods (DAC, weathering) is relatively small, left bars show model-averages of different methods added together, and right bars show bars corrected by model coverage.

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Extended Data Table 1 | Details of scenario databases included in the comparative analysis

	IPCC report name	*Temporal coverage of scenario publication*	*Global scenario count*	*Global model count*	*Source*
SRES	Special Report on Emissions Scenarios	~2000	80	10	(IPCC, 2000)
AR5	5th Assessment report	2010-2014	1184	31	(IPCC, 2014)
SR1.5	Special Report 1.5°C	2015-2018	416	25	(IPCC, 2018)
AR6	6th Assessment report	2015-2021 (includes those in SR1.5)	2304	95	(Byers et al., 2022)